

# COMP7705 Project — Progress Update

**Project Title:** A Multi-Purpose Tool-Using LLM Agent System for End-to-End Task Automation

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## 1. Key Progress and Milestones Achieved

We have completed **Milestones 1–5** (out of 10) ahead of the May 25 deadline:

| # | Milestone                                             | Deadline | Status              |
|---|-------------------------------------------------------|----------|---------------------|
| 1 | System architecture design & proposal                 | Mar 27   | ✅ Completed         |
| 2 | Development environment setup                         | Apr 7    | ✅ Completed         |
| 3 | Core LLM agent execution pipeline                     | Apr 14   | ✅ Completed         |
| 4 | Domain tool integration (Web Search & Python Sandbox) | May 5    | ✅ Completed         |
| 5 | Memory management & tool-use constraints              | May 25   | ✅ Completed (early) |

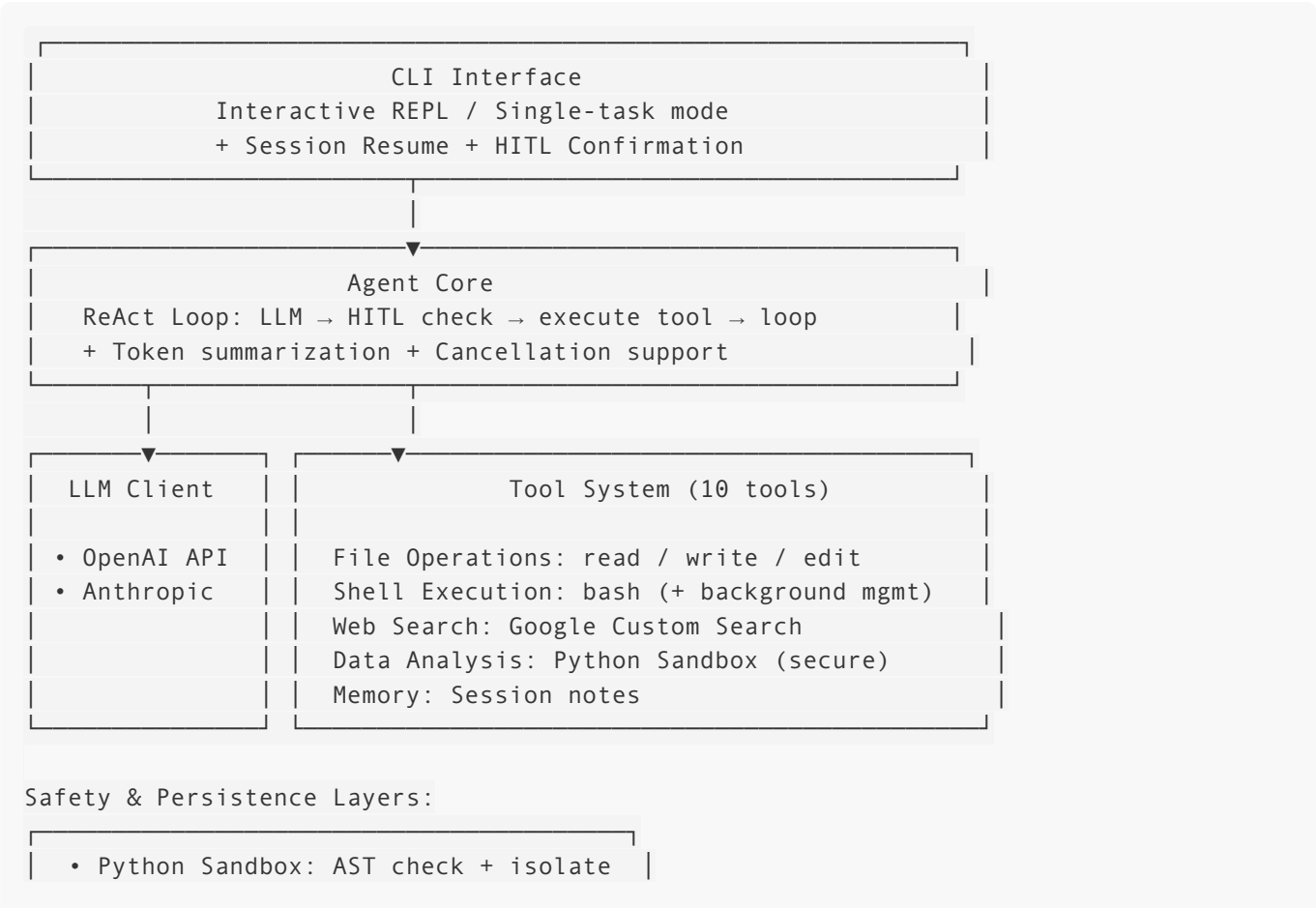
## Technical Highlights

We built a modular LLM agent system from the ground up, consisting of the following core components:

| Module                   | Description                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------------------|
| Agent Execution Pipeline | ReAct-style loop: LLM call → tool selection → execution → feedback → loop                         |
| Multi-LLM Client Layer   | Supports both OpenAI and Anthropic protocols; currently using Claude Opus 4.6 (AWS Bedrock)       |
| Web Search Tool          | Google Custom Search integration for real-time web research                                       |
| Python Sandbox           | Secure code execution with AST-based security checking, module blocklist, file isolation, timeout |
| Human-in-the-Loop (HITL) | Safety constraints that detect dangerous operations and require user confirmation                 |
| Session Persistence      | Multi-session history with save/restore/list, enabling conversation continuity across restarts    |
| Token Management         | Automatic context summarization when approaching token limits                                     |

**Total codebase:** ~3500 lines of Python (excluding tests and documentation).

## 2. System Architecture



- HITL: Regex detection + confirm
- Token Summarization: Auto-compress
- Session Manager: Multi-history

## 3. Feature Demonstrations

### 3.1 Agent Startup — All Tools Loaded Successfully

```
cyq@YUQINGCUI-MB1 Mini-Agent % uv run mini-agent
✓ LLM retry mechanism enabled (max 3 retries)
✓ Loaded Bash Output tool
✓ Loaded Bash Kill tool
Loading Claude Skills...
✓ Discovered 15 Claude Skills
✓ Loaded Skill tool (get_skill)
Loading MCP tools...
MCP timeouts: connect=10.0s, execute=60.0s, sse_read=120.0s
⚠ MCP config file not found: mcp.json

✓ Loaded Bash tool (cwd: /Users/cyq/Code/7705/Mini-Agent)
✓ Loaded file operation tools (workspace: /Users/cyq/Code/7705/Mini-Agent)
✓ Loaded session note tool
✓ Loaded Python sandbox tool (timeout: 30s)
✓ Loaded web search tool (Google Custom Search)
✓ Loaded system prompt (from: /Users/cyq/Code/7705/Mini-Agent/mini_agent/config/system_prompt.md)
✓ Injected 15 skills metadata into system prompt
```

 **Mini Agent – Multi-turn Interactive Session**

Session Info

Model: api\_aws\_third\_anthropic.claude-opus-4-6-v1  
Workspace: /Users/cyq/Code/7705/Mini-Agent  
Message History: 1 messages  
Available Tools: 10 tools

Type /help for help, /exit to quit

You > 

### 3.2 Web Search — Multi-step Task Execution

The agent autonomously searches the web, synthesizes results, and writes output files.

```

You > Search the web for "LLM agent framework 2026" and save top 3 results to test_playground/results.md

Agent > Thinking... (Esc to cancel)

Log file: /Users/cyq/.mini-agent/log/agent_run_20260504_023638.log

Step 1/100

Assistant:

I'll search the web and save the results simultaneously. Let me start by searching and ensuring the output directory exists.

Tool Call: web_search
Arguments:
{
  "num": 3,
  "query": "LLM agent framework 2026"
}
Result: Search results for: "LLM agent framework 2026" (15,900,000 results in 0.35s)

**[1] What is your full AI Agent stack in 2026? : r/AI_Agents - Reddit**
URL: https://www.reddit.com/r/AI_Agents/comments/1rqnv3a/what_is_your_full_ai_agent_stack_in_2026/
Mar 11, 2026 ... Great thread - here's wha...

Tool Call: bash
Arguments:
{
  "command": "mkdir -p test_playground"
}
Result: (no output)

Step 1 completed in 8.15s (total: 8.15s)

Step 2/100

Assistant:

Now I'll save the top 3 results to a markdown file.

Tool Call: write_file
Arguments:
{
  "content": "# Search Results: \"LLM agent framework 2026\"\n\n> Top 3 results from web search\n\n---\n\n# 1. What is your full AI Agent stack in 2026? : r/AI_Agents - Reddit\n\n- **URL**: https://www.reddit.com/r/AI_Agent...\",\n  "path": "test_playground/results.md"
}
Result: Successfully wrote to /Users/cyq/Code/7705/Mini-Agent/test_playground/results.md

Step 2 completed in 9.35s (total: 17.50s)

Step 3/100

Assistant:

Done! I've searched for "LLM agent framework 2026" and saved the top 3 results to `test_playground/results.md`. Here's a summary of what was found:

| # | Title | Source | Date |
|---|---|---|---|
| 1 | What is your full AI Agent stack in 2026? | Reddit (r/AI_Agents) | Mar 11, 2026 |
| 2 | Bridging natural language and GIS: a multi-agent framework for LLM... | Taylor & Francis Online | Feb 23, 2026 |
| 3 | What are the best AI agent builders in 2026? | Reddit (r/automation) | Feb 23, 2026 |

The file is formatted in clean Markdown with URLs, dates, and snippets for each result.

Step 3 completed in 7.17s (total: 24.66s)

```

### 3.3 Python Sandbox — Secure Data Analysis

pandas DataFrame creation and statistical analysis, executed within a sandboxed environment.

You > Use python\_sandbox to create a pandas DataFrame with 5 students and print the average score

Agent > Thinking... (Esc to cancel)

Log file: /Users/cyq/.mini-agent/log/agent\_run\_20260504\_023805.log

Step 1/100

Tool Call: python\_sandbox

Arguments:  
{  
 "code": "import pandas as pd\n\n# Create a DataFrame with 5 students\ndf = pd.DataFrame({\n 'Name': ['Alice', 'Bob', 'Charlie', 'Diana', 'Eve'],\n 'Score': [92, 85, 78, 95, 88]\n})\n\nprint(\"Student DataFrame:\")..."

Result: Student DataFrame:

|   | Name    | Score |
|---|---------|-------|
| 0 | Alice   | 92    |
| 1 | Bob     | 85    |
| 2 | Charlie | 78    |
| 3 | Diana   | 95    |
| 4 | Eve     | 88    |

Average Score: 87.6

Step 1 completed in 6.53s (total: 6.53s)

Step 2/100

Assistant:

Here's the result:

| # | Name    | Score |
|---|---------|-------|
| 0 | Alice   | 92    |
| 1 | Bob     | 85    |
| 2 | Charlie | 78    |
| 3 | Diana   | 95    |
| 4 | Eve     | 88    |

\*\*\*Average Score: 87.6\*\*\*

The DataFrame was created with 5 students and their scores, and the average was calculated using `df['Score'].mean()`. Diana leads with 95, while Charlie has the lowest at 78.

Step 2 completed in 6.16s (total: 12.69s)

### 3.4 Sandbox Security — Dangerous Import Blocked

The AST checker intercepts `import subprocess` before execution, preventing unsafe operations.

You > Use python\_sandbox to run: import subprocess; subprocess.run(['ls'])

Agent > Thinking... (Esc to cancel)

Log file: /Users/cyq/.mini-agent/log/agent\_run\_20260504\_023914.log

Step 1/100

Tool Call: python\_sandbox

Arguments:  
{  
 "code": "import subprocess; subprocess.run(['ls'])"  
}

Error: Security check failed:  
- Blocked import: 'subprocess' (security restriction)

Step 1 completed in 6.20s (total: 6.20s)

Step 2/100

Assistant:

As expected, the `subprocess` module is **blocked** in the Python sandbox. This is by design – the sandbox environment enforces security restrictions that prevent potentially dangerous operations like:

- **subprocess** – running system commands
- **os.system** – executing shell commands
- **Network access** – making external requests
- **File deletion** – removing files from the system

The sandbox is intended for safe **data analysis and computation** (pandas, numpy, math, etc.). If you need to run system commands like `ls`, use the **bash** tool instead:

Tool Call: bash

Arguments:  
{  
 "command": "ls"  
}

Result: CODE\_OF\_CONDUCT\_CN.md

CODE\_OF\_CONDUCT.md

CONTRIBUTING\_CN.md

CONTRIBUTING.md

docs

examples

LICENSE

MANIFEST.in

mini\_agent

mini\_agent.egg-info

pyproject.toml

README\_CN\_minimax.md

README\_CN.md

README.md

scripts

test\_playground

tests

uv.lock

Step 2 completed in 7.65s (total: 13.85s)

Step 3/100

Assistant:

The `bash` tool successfully listed the directory contents. Use **python\_sandbox** for computation/analysis and **bash** for system operations.

Step 3 completed in 5.69s (total: 19.54s)

## 3.5 Session Statistics

You > /stats

### Session Statistics:

Session Duration: 00:06:24

Total Messages: 18

– User Messages: 3

– Assistant Replies: 8

– Tool Calls: 6

Available Tools: 10

API Tokens Used: 6,518

### 3.6 Human-in-the-Loop — Dangerous Command Intercepted

The HITL module detects `sudo apt-get update` and prompts the user for confirmation. After denial, the agent explains why it was blocked and suggests safe alternatives.

```
You > Run bash command: sudo apt-get update

Agent > Thinking... (Esc to cancel)
Log file: /Users/cyq/.mini-agent/log/agent_run_20260522_193548.log

Step 1/100

Tool Call: bash
Arguments:
{
  "command": "sudo apt-get update",
  "timeout": 120
}

Human Confirmation Required

Tool: bash
Risk: Potentially dangerous command detected: sudo apt-get update
[y] Allow [n] Deny [a] Allow all (disable HITL for session)
[y/n/a] > n
Denied: Potentially dangerous command detected: sudo apt-get update

Step 1 completed in 8.58s (total: 8.58s)

Step 2/100

Assistant:
The command `sudo apt-get update` was blocked because it's a potentially dangerous system-level operation that requires elevated privileges (`sudo`).

**Why it was blocked:**
- It modifies system package lists at the OS level
- It requires root/administrator access
- It could affect the stability of the host system

**Notes:**
- It appears you're on macOS (`/Users/cyq/...`), where `apt-get` isn't available anyway (that's a Debian/Ubuntu package manager)
- On macOS, the equivalent would be `brew update` if you're using Homebrew

Would you like me to try an alternative, or is there something specific you're trying to accomplish?

Step 2 completed in 9.69s (total: 18.27s)

You > 
```

### 3.7 Session Resume — Restore Previous Conversations

On restart, the system lists previous sessions. The user selects one to resume, and the full conversation context is restored.

```
Found 4 previous session(s):
[1] 2026-05-22 19:36 (5 msgs) - Run bash command: sudo apt-get update
[2] 2026-05-22 19:14 (5 msgs) - which base model do you use?
[3] 2026-05-22 19:10 (3 msgs) - which base model do you use?
[4] 2026-05-22 19:02 (3 msgs) - who are you?
[n] Start new session

Resume? [1-4/n] > 1
✔ Session restored (4 messages loaded)

Previous conversation:
You: Run bash command: sudo apt-get update
Agent: The command `sudo apt-get update` was blocked because it's a potentially dangerous...
```

#### 🤖 Mini Agent - Multi-turn Interactive Session

##### Session Info

Model: api\_aws\_third\_anthropic.claude-opus-4-6-v1  
Workspace: /Users/cyq/Code/7705/Mini-Agent  
Message History: 5 messages  
Available Tools: 10 tools

Type `/help` for help, `/exit` to quit

You > █

## 4. Major Challenges and Issues

| Challenge                                                                                                          | How We Resolved It                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Platform API uses non-standard authentication format                                                               | Implemented custom auth header construction with auto-detection logic                                 |
| matplotlib crashes sandbox on every call due to backend initialization                                             | Pre-import at tool initialization time with non-interactive <code>Agg</code> backend                  |
| Agent bypasses HITL by using equivalent commands (e.g., <code>find -delete</code> instead of <code>rm -rf</code> ) | Expanded pattern list; identified as key motivation for Self-verification module                      |
| Terminal input conflicts between Esc-key cancellation and HITL confirmation                                        | Implemented pause/resume mechanism using <code>termios</code> to temporarily release terminal control |
| Token accumulation during long sessions                                                                            | Automatic summarization compresses execution history while preserving user intent                     |

## 5. Planned Milestones and Focus for Coming Weeks

| Milestone                          | Deadline | Key Deliverables                                                                       |
|------------------------------------|----------|----------------------------------------------------------------------------------------|
| 6. Interim Report & Presentation   | June 1   | 20-min Zoom presentation with live demo; Interim Report draft                          |
| 7. Self-verification module        | June 16  | LLM-based output verification; intent-tracking to prevent safety bypasses              |
| 8. Benchmarking & ablation studies | July 6   | Multi-domain test suite (web research, data analysis, coding); quantitative evaluation |
| 9. Final source code & Webpage     | July 13  | Code cleanup, documentation, Project Webpage update                                    |
| 10. Final Report & Oral Exam       | July 17  | ~15,000-word report; presentation slides; oral examination                             |

### Detailed Plan

**Week of May 25 – June 1 (Interim Presentation):** - Prepare 20-minute presentation slides covering architecture, implementation, and live demo - Draft Interim Report (system design, methodology, preliminary results) - Update Project Webpage with current progress

**June 2 – June 16 (Self-verification):** - Design and implement Self-verification module — use a secondary LLM call to validate Agent outputs - Implement intent-tracking: when user denies an operation, record the intent and block semantically equivalent attempts - Explore LLM-as-Judge approach for assessing task completion quality

**June 17 – July 6 (Evaluation & Benchmarking):** - Construct custom benchmark with 30+ representative tasks across three domains: - Web Research (e.g., "Find and compare the top 3 LLM frameworks") - Data Analysis (e.g., "Load this CSV, compute statistics, generate a chart") - Coding (e.g., "Create a Python script that does X, test it, fix bugs") - Define evaluation metrics: task success rate, token cost, execution time, safety violation rate - Conduct ablation studies: - With vs. without Python Sandbox (security impact) - With vs. without HITL (safety vs. autonomy trade-off) - With vs. without Self-verification (correctness improvement) - With vs. without Session Resume (context continuity impact)

**July 7 – July 17 (Final Report & Exam):** - Finalize source code and documentation - Write Final Report (~15,000 words) covering literature review, methodology, implementation, evaluation results, and conclusion - Prepare oral examination presentation and demo

## 6. Individual Contributions

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All members participated in system design discussions, code review, testing, and documentation throughout the project. Below highlights each member's primary focus areas:

| Member                 | Primary Focus                                                                             |
|------------------------|-------------------------------------------------------------------------------------------|
| Cui Yuqing<br>(Leader) | System architecture, core pipeline implementation, tool development, project coordination |
| Zhang Zhiyu            | System architecture, core implementation, Project Webpage maintenance                     |
| Kuang Yang             | System architecture, tool integration, API protocol research                              |
| Huang Danqi            | System architecture, Python Sandbox security design, evaluation framework, testing        |
| Wang Junjie            | System architecture, memory management module, documentation, literature review           |

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## 7. Questions

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1. We are available for the interim presentation in the week of June 1 via Zoom. Could you please let us know the specific date and time when arranged?
  2. Regarding the Final Report, should we include detailed code-level explanations of each module, or focus more on the high-level design decisions and experimental results?
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**Project Webpage:** <https://wp2025.cs.hku.hk/msp25023/>

**GitHub (Progress Updates):** <https://github.com/Peter-Quest/COMP7705-Progress-Updates>